

Motivation for the RCP framework – the organizing framework for my research

The goal of my research is to understand the computations that support **natural behavior**. Although such behavior appears complex, a normative viewpoint allows us to ask a simpler question: *What problems must the brain solve to behave adaptively in the real world?*

A first challenge is that the brain must choose actions that maximize utility despite receiving only *impoverished, noisy observations*. The normative solution is given by **Partially Observable Markov Decision Processes (POMDPs)**, which unify three coupled computations:

- i. **Perception** – inferring latent states from uncertain observations,
- ii. **Learning** – updating internal models from feedback, and
- iii. **Planning** – selecting actions that maximize long-term reward.

But classical POMDPs are insufficient because both the environment and the agent's goals *change over time*. Flexible behavior requires **contextual inference**: continuously inferring which internal model or context is currently active. Moreover, exact Bayesian inference and planning become intractable in natural settings, forcing the brain to trade off *optimality* with *computational cost*, as formalized by **resource rationality**.

These ingredients together define the **Resource-rational Context-dependent POMDP (RCP)** framework, which explains how perception, learning, planning, contextual inference, and computational constraints jointly give rise to natural behavior.

My prior work has investigated each of these components in depth. I have developed **normative theoretical models** of these computations and tested them using **human behavior**. I have also derived **neural predictions**, examined **clinical populations** to understand how these computations break down, and proposed **machine-learning analogs** of the algorithms implied by the RCP framework.

Looking forward, my lab will build on this foundation by studying the **pairwise interactions** between perception, learning, and planning as stepping stones toward an integrated account of how all three computations jointly support natural behavior in dynamic environments.

Below, I present my publications organized according to this structure (in the same format as explained in the research proposal), indicating where each contribution lies within the RCP framework:

S1 Shivkumar, S., DeAngelis, G. C., & Haefner, R. M. (2025).

Hierarchical motion perception as causal inference. Nature Communications, 16(1), 3868. (Link)

RCP components: **perception**, **contextual inference**, **resource rationality**, **theory**, **behavior**

Summary: This work presents a hierarchical Bayesian model, constrained by human psychophysics, where motion perception is solved via causal inference over latent reference frames. The model unifies motion integration and segmentation by dynamically inferring world structure, supporting a normative explanation for key Gestalt principles.

S2 Noel, J. P.^{*}, Shivkumar, S.^{*}, Dokka, K.^{*}, Haefner, R. M.[†], & Angelaki, D.[†] (2022).

Aberrant causal inference and a compensatory mechanism in Autism Spectrum Disorder. eLife, 11. (Link)

RCP components: **perception**, **contextual inference**, **behavior**, **clinical prediction**

Summary: This study found that individuals with Autism Spectrum Disorder (ASD) exhibit an increased implicit prior probability of combining auditory and visual cues during spatial localization tasks/ Conversely, explicit reports of common cause were decreased, suggesting a compensatory choice bias mechanism used by ASD individuals to counteract their implicit bias toward integration.

S3 Shivkumar, S., Cappelloni, M. S., Maddox, R. K., & Haefner, R. M. (under review, eLife).

Inferring sources of suboptimality in perceptual decision making. (Link)

RCP components: **perception**, **resource rationality**, **theory**, **behavior**

Summary: This research formalizes different sources of suboptimality in perceptual decision-making, such as biases and approximate computations (finite sampling), within the ideal observer framework. It uses a causal inference task involving choice-uninformative visual cues to empirically dissociate these sources, which cannot be separated using data from traditional binary discrimination tasks

S4 Shivkumar, S., Fisher, V. L., Shadlen, M. N., & Wolpert, D. M. (manuscript under preparation).

Multiple sources of evidence are multiplexed to make a single decision. (Link)

RCP components: **perception**, **contextual inference**, **theory**, **behavior**

Summary: This study modeled multisensory evidence integration in a dual cue (motion and color) decision task, comparing various decision architectures. The behavioral data were best fit by a unweighted parallel, single-accumulator model, suggesting that evidence sources are combined without considering the per-trial reliability of color/motion cues.

S5 Penaloza, B., Shivkumar, S., Lengyel, G., DeAngelis, G. C., & Haefner, R. M. (2024).

Causal inference predicts the transition from integration to segmentation in motion perception. Scientific Reports. (Link)

RCP components: **perception**, **contextual inference**, **behavior**

Summary: This work tests the key prediction that the switch between motion integration and segmentation is governed by

Bayesian Causal Inference (CI). Experimental results confirm that the perceptual transition point from integration to segmentation, along with the point of maximum perceptual variability, shifts dependent on sensory uncertainty.

- S6** Cappelloni, M. S., **Shivkumar, S.**, Haefner, R. M., & Maddox, R. K. (2019).

Task-uninformative visual stimuli improve auditory spatial discrimination in humans. PLOS ONE. ([Link](#))

RCP components: **perception**, **contextual inference**, **resource rationality**, **behavior**

Summary: This study examined the effect of spatially aligned, task-uninformative visual stimuli on human auditory spatial discrimination. Human performance improved with these visual cues, demonstrating an effect not predicted by the ideal Bayesian observer, whose response is mathematically unaffected by the visual cues.

- S7** Dong, Y., Lengyel, G., **Shivkumar, S.**, Anzai, A., DiRisio, G. F., Haefner, R. M., & DeAngelis, G. C. (2025).

Rewarding animals based on their subjective percepts. PLoS Biology, 23(5). ([Link](#))

RCP components: **perception**, **behavior**

Summary: This study introduces a Bayesian online method to infer animals' moment-to-moment perceptual biases directly from behavior, enabling reward delivery contingent on subjective percepts rather than objective stimuli.

- S8** **Shivkumar, S.**, Naik, A. S., Ingram, J. N., Lengyel, M.[†], & Wolpert, D. M.[†] (manuscript under preparation, presented as a talk at MLMC – SfN 2024).

Contributions of visual, dynamic, and kinematic cues to compositional motor learning. ([Link](#))

RCP components: **learning**, **contextual inference**, **theory**, **behavior**

Summary: This study investigated compositional motor learning and generalization of motor memories using a 3-D robotic reaching task. Results showed participants infer partially compositional representations (around 1/3 proportion) from visual, dynamic, or kinematic cues, supporting structured generalization.

- S9** **Shivkumar, S.**, Lengyel, M.[†], & Wolpert, D. M.[†] (2025).

Curriculum effects in multitask learning through the lens of contextual inference. Current Opinion in Neurobiology. ([Link](#))

RCP components: **learning**, **contextual inference**, **theory**, **machine learning**

Summary: This review proposes contextual inference (COIN) as a unified framework to explain diverse curriculum effects (e.g., acquisition speed vs. long-term retention) observed across motor, perceptual, and machine learning domains/ The framework shows that retention patterns are determined by the interplay between the history-to-cue weighting ratio and feedback precision, resolving the puzzle of contradictory curriculum outcomes.

- S10** **Shivkumar, S.**, Wolpert, D. M.[†], & Lengyel, M.[†] (manuscript under preparation).

Contextual inference: a unified explanation for curriculum effects. ([Link](#))

RCP components: **learning**, **contextual inference**, **theory**, **behavior**

Summary: This work uses the Contextual Inference (COIN) model to provide a unified explanation for curriculum effects observed in motor learning paradigms. The model predicts that blocked training leads to increased reliance on stimulus history over sensory cues, resulting in poorer retention when tested in an random curriculum.

- S11** **Shivkumar, S.**, Ingram, J. N., Lengyel, M.[†], & Wolpert, D. M.[†] (manuscript under preparation; presented in Cosyne 2025).

Contextual inference accounts for differences in motor learning under distinct curricula. ([Link](#))

RCP components: **learning**, **contextual inference**, **theory**, **behavior**

Summary: This research applies the Contextual Inference framework to explain how different training schedules (curricula) affect motor learning outcomes. A key finding and hypothesis is that interleaving null trials within a blocked curriculum improves retention by reducing the predictability based on task history, which is verified using behavioral data.

- S12** Cesanek, E., **Shivkumar, S.**, Ingram, J. N., & Wolpert, D. M. (2024).

Ouvrai opens access to remote virtual reality studies of human behavioural neuroscience. Nature Human Behaviour. ([Link](#))

RCP components: **learning**, **behavior**

Summary: This resource introduces Ouvrai, an open-source, end-to-end software solution facilitating the design and execution of remote Virtual Reality (VR) studies for human behavioral neuroscience. The platform was validated by successfully replicating established motor learning paradigms, including spontaneous recovery and dual adaptation, using remote VR headsets.

- S13** Chatteraj, A.*, **Shivkumar, S.***, Ra, Y., & Haefner, R. M. (2021).

A confirmation bias due to approximate active inference. CogSci Proceedings. ([Link](#))

RCP components: **planning**, **resource rationality**, **behavior**

Summary: This work describes a confirmation bias in active sensing, where eye movements are biased by current beliefs when gathering new information for perceptual judgments.

- S14** Chatteraj, A.*, **Shivkumar, S.***, Ra, Y., & Haefner, R. M. (manuscript under preparation).

Approximate inference drives biased active evidence accumulation. ([Link](#))

RCP components: **planning**, **resource rationality**, **theory**, **behavior**

Summary: This work develops an active evidence integration model for tasks where eye movements are required to gather new information. Confirmation bias behavior observed in this task is modeled by a computational model that uses approximate computations (sampling) within an optimal Bayesian active sensing strategy.

- S15 Shivkumar, S., Muralidharan, V., & Chakravarthy, V. S. (2017).**
A biologically plausible architecture of the striatum for context-dependent reinforcement learning. Frontiers in Neural Circuits. ([Link](#))
RCP components: **planning, contextual inference, theory**
Summary: This paper proposes a biologically plausible network model of the basal ganglia (BG) microanatomy – exploiting the modularity of the striatum within a modular Reinforcement Learning (RL) framework to efficiently solve multi-context, context-dependent RL tasks.
- S16 Lengyel, G. *, Shivkumar, S. *, & Haefner, R. M. (2023).**
A general method for testing Bayesian models using neural data. UniReps Workshop. ([Link](#))
RCP components: **perception, neural predictions, theory**
Summary: This paper proposes a general method for testing Bayesian models against neural data by mapping model-derived beliefs (posteriors) onto neural activity.
- S17 Shivkumar, S. *, Lange, R. *, Chattoraj, A. *, & Haefner, R. (2018).**
A probabilistic population code based on neural samples. NeurlPS. ([Link](#))
RCP components: **perception, resource rationality, neural predictions, theory**
Summary: This work demonstrates a link between neural sampling models (Bayesian encoding) and Probabilistic Population Codes (PPCs, Bayesian decoding) within a linear Gaussian model. It shows that samples drawn from a Bayesian encoding model approximate an exponential family with linear sufficient statistics, providing a way to derive the decoding model implied by the encoding model.
- S18 Lengyel, G. *, Shivkumar, S. *, & Haefner, R. M. (2025).**
Bayesian causal inference unifies perceptual and neuronal processing of center-surround motion in MT. bioRxiv. ([Link](#))
RCP components: **perception, neural predictions, theory**
Summary: This study uses Bayesian causal inference over latent reference frames to unify perceptual and neuronal processing of center-surround motion, specifically in area MT. It shows that the diverse, seemingly contradictory response properties of MT neurons can be understood as manifestations of the single underlying computation of inferring causal structure.
- S19 Lange, R. D., Shivkumar, S. *, Chattoraj, A. *, & Haefner, R. M. (2023).**
Bayesian encoding and decoding as distinct perspectives on neural coding. Nature neuroscience, 26(12), 2063-2072. ([Link](#))
RCP components: **perception, resource rationality, neural predictions, theory**
Summary: This perspective distinguishes Bayesian encoding (inference in an internal generative model) and Bayesian decoding (recovering stimulus information from neural activity).
- S20 Levenstein, D., Alvarez, V. A., Amarasingham, A., Shivkumar, S., et al. (2023).**
On the role of theory and modeling in neuroscience. Journal of Neuroscience, 43(7). ([Link](#))
RCP components: **theory, neural predictions**
Summary: This perspective piece advocates for a pragmatic framework to guide neuroscience, arguing that theories and models must integrate descriptive ("what"), mechanistic ("how"), and normative ("why") explanations.